

# **SignSight: An AI-Driven Web Platform for Tamil Sign Language Recognition, Learning Evaluation, and Emotion-Aware Accessibility**

25-26J-404



B.Sc. (Hons) Degree in Information Technology Specializing in  
Information Technology

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Sri Lanka

August 2025

## **1. Project Title**

SignSight: An AI-Driven Web Platform for Tamil Sign Language Recognition, Learning Evaluation, and Emotion-Aware Accessibility

## **2. Project Description**

SignSight is an AI-driven web-based platform designed to improve communication and learning for hearing-impaired individuals. The system integrates multiple intelligent components including sign language recognition, speech-to-sign conversion, learning evaluation, behaviour analysis, and emotion-aware support.

## **3. Project Objectives**

- 1 Develop a Tamil Audio/Video to Sign Language conversion system
- 2 Implement real-time sign detection using computer vision
- 3 Design a learning evaluation system with performance tracking
- 4 Incorporate behaviour analysis for attention monitoring
- 5 Develop an emotion-aware module for child-focused learning

## **4. Project Scope**

### **In Scope:**

- 1 Audio/Video to Sign Language conversion
- 2 Real-time sign detection
- 3 Learning evaluation system
- 4 Behaviour and emotion analysis
- 5 Web-based system with dashboards

### **Out of Scope:**

- 1 Full sentence-level sign translation
- 2 Support for multiple languages
- 3 Hardware-based gesture devices

## **5. Key Deliverables**

- 1 Machine Learning Models for all components
- 2 Integrated Web Application
- 3 User Dashboard and Reporting System
- 4 Final Research Paper (IEEE format)
- 5 Testing and Evaluation Reports

## **6. Stakeholders**

- 1 Hearing-impaired individuals
- 2 Students learning Tamil Sign Language
- 3 Parents and educators
- 4 Academic supervisors
- 5 Research community

## **7. Technologies Used**

- 1 Machine Learning: CNN, LSTM, GRU, MobileNetV2
- 2 Computer Vision: MediaPipe, OpenCV
- 3 Speech Processing: Speech-to-Text models
- 4 Frontend: React.js
- 5 Backend: Node.js / Python
- 6 Database: MongoDB / MySQL

## **8. Project Timeline**

- 1 Requirement Analysis
- 2 Data Collection
- 3 Model Development
- 4 System Integration
- 5 Testing and Evaluation
- 6 Final Deployment and Documentation

## **9. Key Risks**

- 1 Limited Tamil Sign Language datasets
- 2 Accuracy affected by real-world conditions
- 3 Integration complexity of multiple AI components
- 4 Performance challenges in real-time processing

## **10. Success Criteria**

- 1 High accuracy in recognition and conversion
- 2 Successful integration of all components
- 3 Effective learning evaluation and feedback
- 4 Improved accessibility for users

## **11. Expected Impact**

- 1 Enhances communication for deaf community
- 2 Supports Tamil Sign Language education
- 3 Provides inclusive AI-based learning platform
- 4 Contributes to research in low-resource languages